Shelby Dixon

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Computer Engineering Candidate | Applied AI, Data Validation, Automation & Edge Sensing

Computer Engineering student with hands-on experience building AI/data validation pipelines, backend tools, and edge-sensing prototypes for agriculture, biosecurity, public-sector, and infrastructure problems. Brings 10+ years of farm/field operations experience plus Python, SQL, FastAPI, Google Cloud, RAG, embedded systems, and stakeholder-facing communication.

EDUCATION

University of Hawai'i at Manoa, Department of Electrical & Computer Engineering | B.S. Computer Engineering, Dec 2026
Coursework: Machine Learning, Algorithms, Object-Oriented Programming, Digital Design, Signals & Systems, Microelectronics, Circuit Analysis, Robotics & Cyberphysical Systems

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, JavaScript/HTML/CSS, PowerShell, MATLAB, Verilog/VHDL
Cloud/Tools: FastAPI, BigQuery, Google Cloud Run, Git, Jupyter, Apify, Excel automation
AI/Data: data pipelines, data cleaning/modeling, JSON/schema export, visualization, RAG, embeddings, FAISS, PyTorch
Hardware: Raspberry Pi, FPGA/Vivado, sensors, oscilloscopes, DMMs, circuit simulation, soldering

EXPERIENCE

Oceanit | Software Engineering Intern | Honolulu, HI | Jun 2026 - Present
- Develop Python/PowerShell tooling for AI-assisted document and data workflows supporting DOT airport lease-review and crash-analysis initiatives.
- Built structured JSON export and QA checks for Excel review workbooks, including citations/page arrays, score changes, null semantics, path validation, and SHA-256 source-file verification.
- Helped frame fatal-crash dashboard analysis by reducing broad question sets into themes, filters, visuals, and user-facing vs. internal checks.

University of Hawai'i Government Relations | Business & Technical Lead, AI for Legislative Intelligence | Aug 2025 - Jan 2026
- Co-led AI platform to monitor Hawai'i Legislature hearings, bills, testimony, and vote records for UH-relevant content.
- Built Python/BigQuery/Apify/GCP workflows and FastAPI/Cloud Run services for transcript ingestion, indexing, semantic search/RAG, keyword detection, and bill-version comparison.

Institute for Human Services, Taking Root Program | Urban Agriculture Specialist / Project Manager | Mar 2020 - Aug 2024
- Led greenhouse, hydroponic, and soil-based production operations from setup through daily workflows, maintenance, supplies, and partner coordination.
- Applied farm operations experience to systems thinking, resource planning, and practical technology needs in agriculture/community settings.

SELECTED PROJECTS & RESEARCH

PUALEI | Low-Power AI Plant-Health Monitoring | UH Manoa | 2025 - Present
- Develop edge-AI and sensor concept for space and resilient agriculture using plant-stress sensing, capillary-wick water delivery, and onboard inference.
- Coordinate interdisciplinary build planning and translate system goals into milestones, stakeholder materials, and research direction.

TRACE-CRB | Acoustic Field Triage Prototype | Oceanit/UH | 2026
- Built early contact-acoustic/vibration sensing prototype for coconut rhinoceros beetle inspection triage: "listen before you dig."
- Developed Raspberry Pi/touchscreen demo converting signal activity into simple field-facing outputs: Inspect First, Follow Up, or Clear.

VENOM-RAG | RAG Poisoning and Hidden Prompt Injection Demo | UH ECE | Jan 2026 - Present
- Built cybersecurity-focused AI demo showing how poisoned PDFs/hidden text can influence RAG retrieval and model responses using Python, FAISS, embeddings, PDF extraction, and FastAPI/Jupyter tooling.

Hawai'i Space Grant Consortium | Undergraduate Researcher | Sep 2024 - Sep 2025
- Supported NASA-funded student missions including First Nations Launch and RockSat-X through payload design, embedded integration, electrical testing, and flight-readiness preparation.

PUBLICATION

Dixon, S. "Adrian: A Privacy-First Embodied AI Platform with Ethical Design Constraints." Manoa Horizons, Vol. X. hdl.handle.net/10125/112621